Git Workshop

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2026 Semester 2

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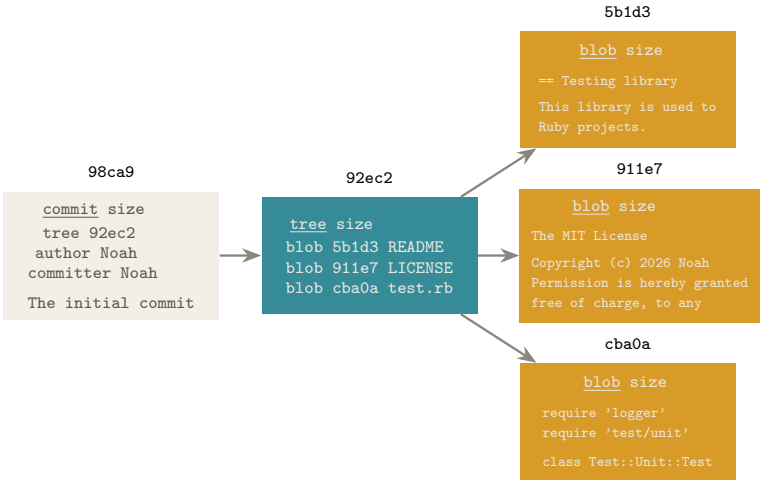
Unlike systems like Google Docs, Git only stores the state of the repository at specific points called **commits**.
When a commit is created Git takes a snapshot of the repository at that point and refers to it using a sha1 hash.

What is a commit?



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This gives us a complete local copy of the repository, including the full history
We can also clone using the ssh protocol instead of https

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git config user.name "Noah"
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If we don't want to repeat this for each repository we can add the '`--global`' flag.

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If we don't want to repeat this for each repository we can add the '`--global`' flag.
We can view the repository or global config using

```
git config --edit --global
```

This can be used to store authentication tokens.

Files

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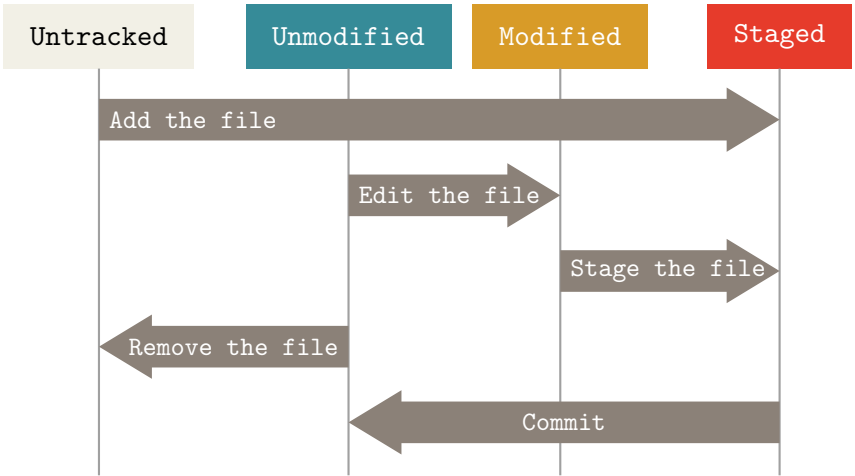
Warning

Don't commit environment files

File areas


Untracked
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Add the file
Edit the file
Stage the file
Remove the file
Commit

Files cont.

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There are numerous useful options such as

- `-m <msg>`
- `-S`

Once we've made this commit we can push it to the remote repo using:

```
git push
```

Remark

Filenames in ".gitignore" are automatically ignored by git.

Removing files

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This doesn't remove the file from the history.

Commit messages

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Figure: Randall Munroe, https://xkcd.com/1296

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Warning

Some classes will mark you down for uninformative commit messages

History

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git log
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Commits are referred to by a sha1 hash, for example

```
d05e2ac13494daf33d26004d52f577bea19c9e52
```

There are various options to modify the output format, including `--pretty`. It is displayed using the Unix command *less*

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HEAD is a pointer the latest commit on the current branch

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- `git commit --amend`
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Remark

Some classes will mark you down if you include junk files in your repository, this is how you will remove them

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```
git reset HEAD~5
```

git revert

Git revert makes a new commit that undo the changes made by a individual commit *without deleting* that commit. The `--no-commit` flag is used to put the changes in the staging area instead of automatically creating a commit.

commit –amand

This command is used to modify the most recent commit, allowing the commit message to be changed and files to be added. However it will completely overwrite the old commit.

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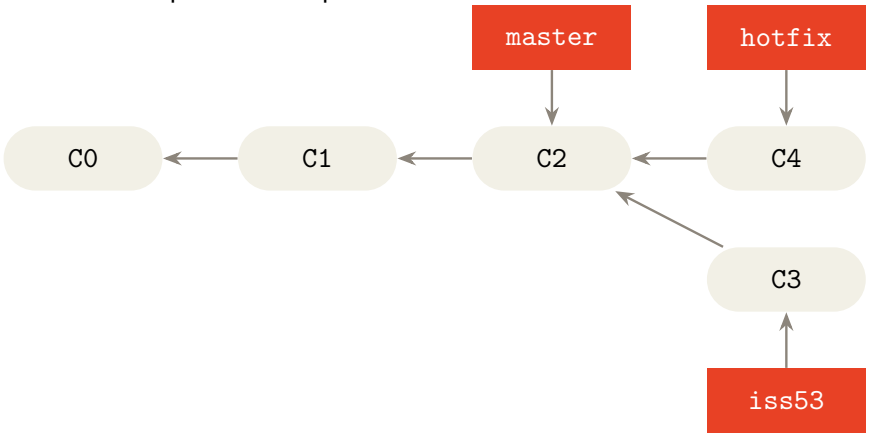
Remark

A detached HEAD means you are not on any branch

Branches

Branches are pointers to specific commits.



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Say we want to merge branches iss53 and hotfix to apply the hotfix to our iss53 branch. We would first switch to the iss53 branch (using git checkout iss53) and then run:

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Once the conflicts are resolved stage the changes and continue the merge using:

```
git merge hotfix --continue
```

PEP

Thanks to the book Pro Git (second edition) by Scott Chacon and Ben Straub.
Thanks to the Honorable Rudransh Kala for something.